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Batch c2 prn 1262252502

Data Structures Assignment 1

Code:

```
#include <stdio.h>

void inputMatrix(int a[10][10], int r, int c)
{
    int i, j;
    for(i = 0; i < r; i++)
        for(j = 0; j < c; j++)
        {
            printf("Enter element [%d][%d]: ", i, j);
            scanf("%d", &a[i][j]);
        }
}

void displayMatrix(int a[10][10], int r, int c)
{
    int i, j;
    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
            printf("%d\t", a[i][j]);
        printf("\n");
    }
}

void addMatrix(int a[10][10], int b[10][10], int r, int c)
{
    int i, j, sum[10][10];
```

```

    for(i = 0; i < r; i++)
        for(j = 0; j < c; j++)
            sum[i][j] = a[i][j] + b[i][j];

    printf("\nAddition of Matrices:\n");
    displayMatrix(sum, r, c);
}

void subtractMatrix(int a[10][10], int b[10][10], int r, int c)
{
    int i, j, sub[10][10];

    for(i = 0; i < r; i++)
        for(j = 0; j < c; j++)
            sub[i][j] = a[i][j] - b[i][j];

    printf("\nSubtraction of Matrices:\n");
    displayMatrix(sub, r, c);
}

void multiplyMatrix(int a[10][10], int b[10][10], int r1, int c1, int c2)
{
    int i, j, k, mul[10][10];

    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c2; j++)
        {
            mul[i][j] = 0;

```

```

        for(k = 0; k < c1; k++)
            mul[i][j] += a[i][k] * b[k][j];
    }
}

printf("\nMultiplication of Matrices:\n");

displayMatrix(mul, r1, c2);
}

void transposeMatrix(int a[10][10], int r, int c)
{
    int i, j, t[10][10];

    for(i = 0; i < r; i++)
        for(j = 0; j < c; j++)
            t[j][i] = a[i][j];

    printf("\nTranspose of Matrix:\n");

    displayMatrix(t, c, r);
}

int main()
{
    int a[10][10], b[10][10];
    int r1, c1, r2, c2, choice;

    printf("Enter rows and columns of Matrix 1: ");
    scanf("%d%d", &r1, &c1);

    printf("Enter elements of Matrix 1:\n");
    inputMatrix(a, r1, c1);

```

```
printf("Enter rows and columns of Matrix 2: ");
```

```
scanf("%d%d", &r2, &c2);
```

```
printf("Enter elements of Matrix 2:\n");
```

```
inputMatrix(b,r2,c2);
```

```
printf("\nEnter your choice: ");
```

```
scanf("%d", &choice);
```

```
switch(choice)
```

```
{
```

```
    case 1:
```

```
        if(r1 == r2 && c1 == c2)
```

```
            addMatrix(a, b, r1, c1);
```

```
        else
```

```
            printf("Addition not possible!");
```

```
        break;
```

```
    case 2:
```

```
        if(r1 == r2 && c1 == c2)
```

```
            subtractMatrix(a, b, r1, c1);
```

```
        else
```

```
            printf("Subtraction not possible!");
```

```
        break;
```

```
    case 3:
```

```
        if(c1 == r2)
```

```
            multiplyMatrix(a, b, r1, c1, c2);
```

```
        else
```

```
        printf("Multiplication not possible!");  
        break;  
  
    case 4:  
        transposeMatrix(a, r1, c1);  
        break;  
    default:  
        printf("Invalid Choice!");  
}  
return 0;}
```

output:

root@DESKTOP-MP3L7BK:~# gedit assignment1.c

^C

root@DESKTOP-MP3L7BK:~# gcc assignment1.c

root@DESKTOP-MP3L7BK:~# ./a.out

Enter rows and columns of Matrix 1: 2 2

Enter elements of Matrix 1:

Enter element [0][0]: 1

Enter element [0][1]: 2

Enter element [1][0]: 3

Enter element [1][1]: 4

Enter rows and columns of Matrix 2: 2 2

Enter elements of Matrix 2:

Enter element [0][0]: 5

Enter element [0][1]: 6

Enter element [1][0]: 7

Enter element [1][1]: 8

Enter your choice: 1

Addition of Matrices:

6 8

10 12

Enter your choice: 2

Subtraction of Matrices:

-4 -4

-4 -4

Enter your choice: 3

Multiplication of Matrices:

19 22

43 50

Enter your choice: 4

Transpose of Matrix:

1 3

2 4

Enter your choice: 5

Invalid Choice!

Assignment photos:





